

Chapter 9

Software Configuration Management

First Law of System Engineering

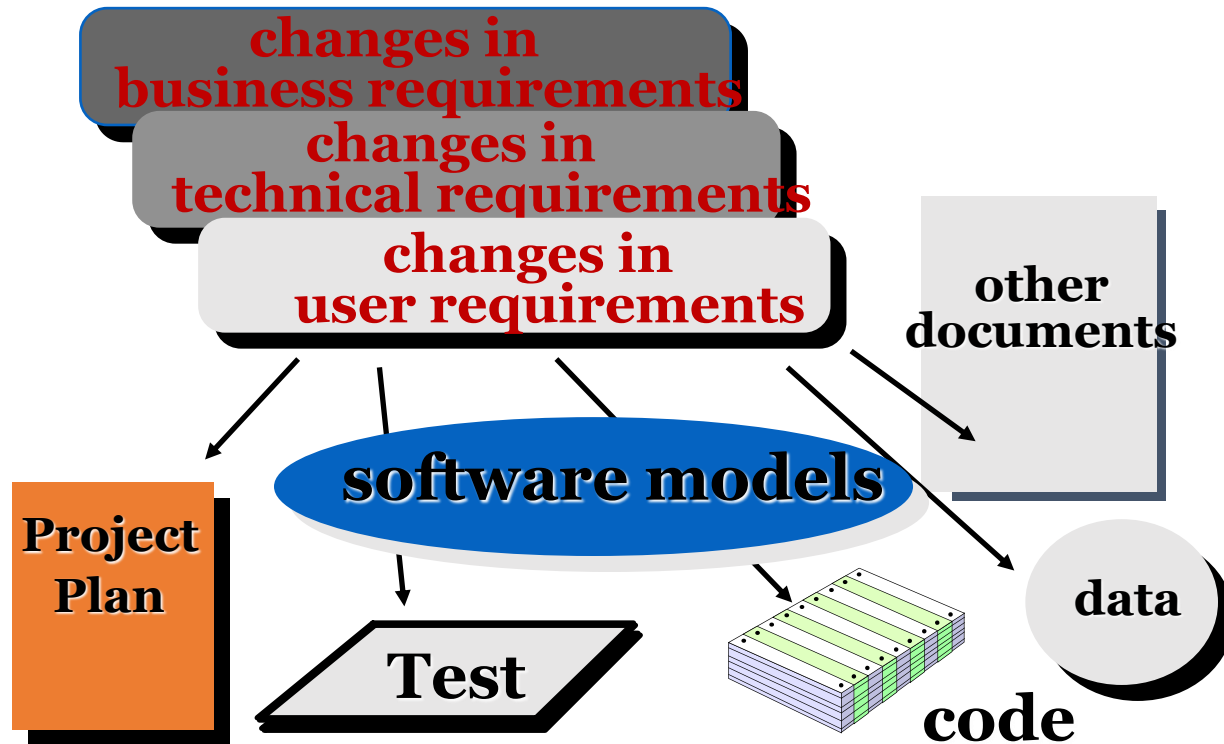
- No matter where you are in the system life cycle, **the system will change, and the desire to change it will persist throughout the life cycle.**

Bersoff *et al.*, 1980

Software Configuration Management (SCM)

- A set of activities that have been developed to manage change throughout the life cycle of computer software
- A software quality assurance activity that is applied throughout the software process
- SCM activities are developed to
 - 1) identify change
 - 2) control change
 - 3) ensure that change is being properly implemented
 - 4) report changes to others who may have an interest

What Are These Changes?



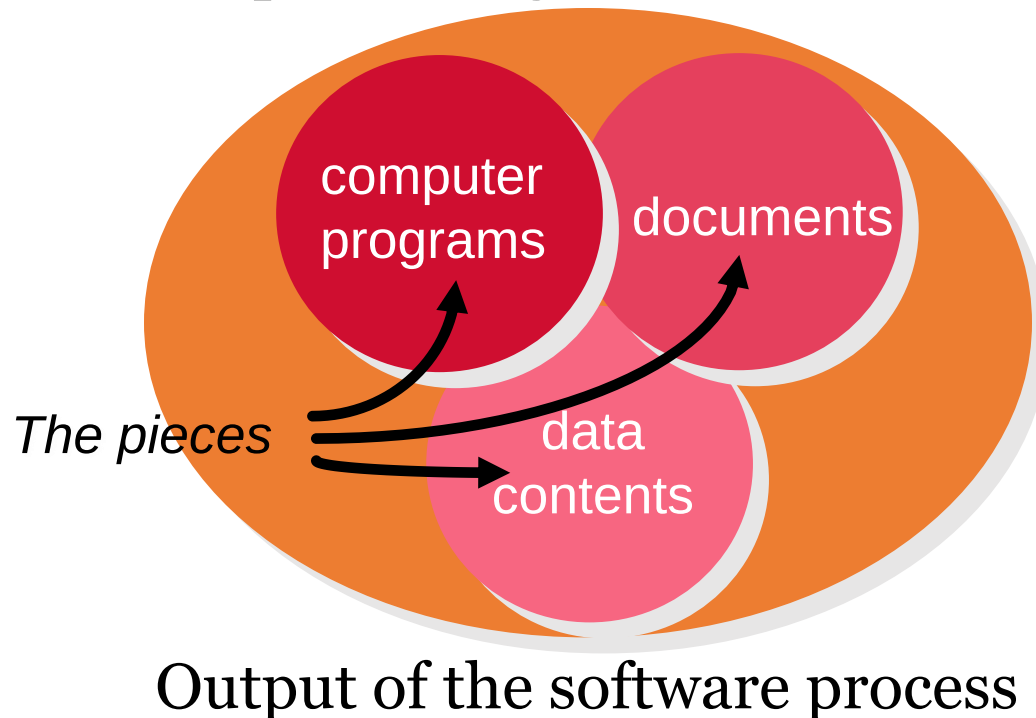
Why SCM?

- Multiple people working in a given project
- Different versions of documents/Software
- Different releases to customers

**Need for coordination &
Streamlined processes**

Software Configuration Item

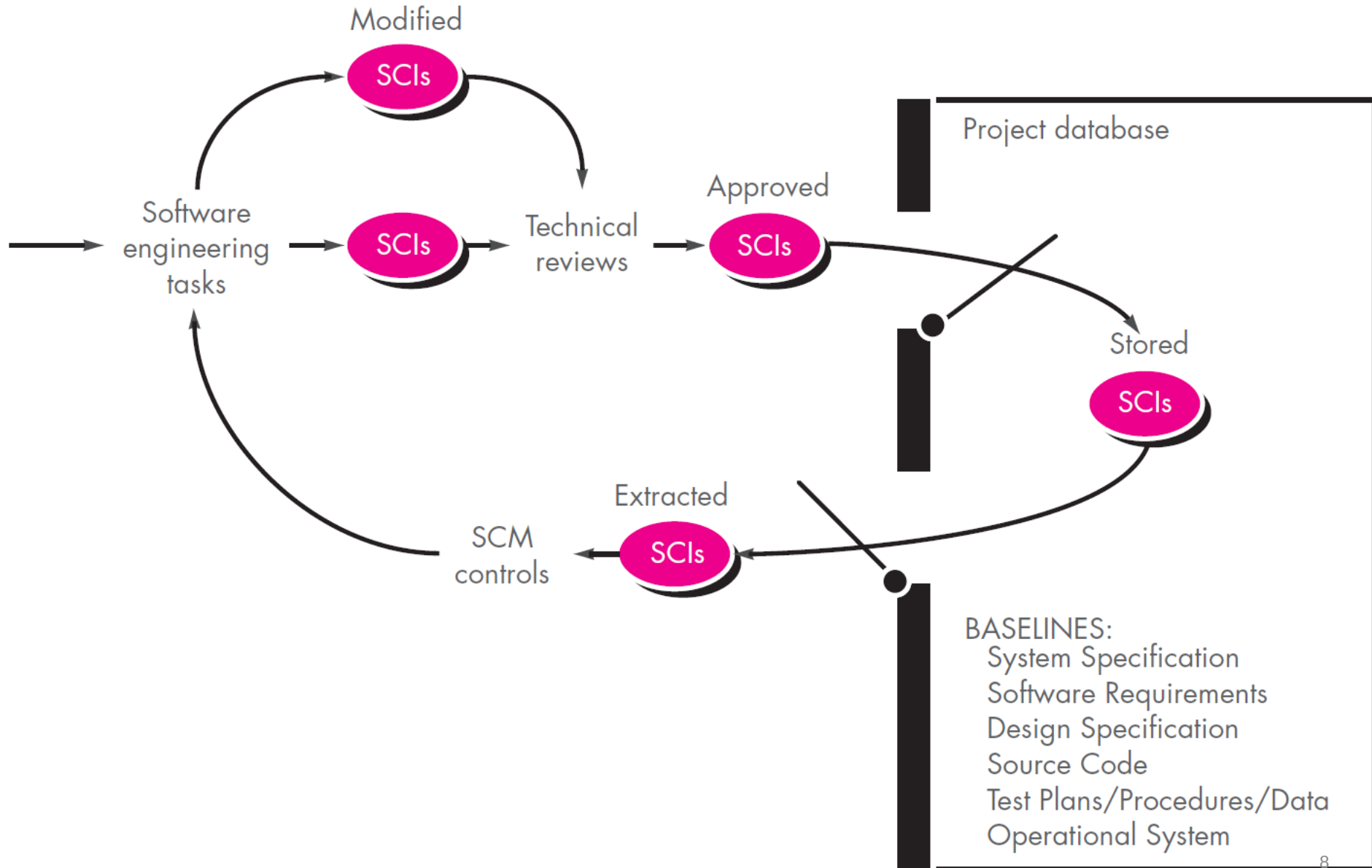
- The items that comprise all information produced as part of the software process are collectively called a *software configuration*.
- An SCI is a named element of information that is created as part of the software engineering process, e.g., a single UML diagram, the complete design document.



Baselines

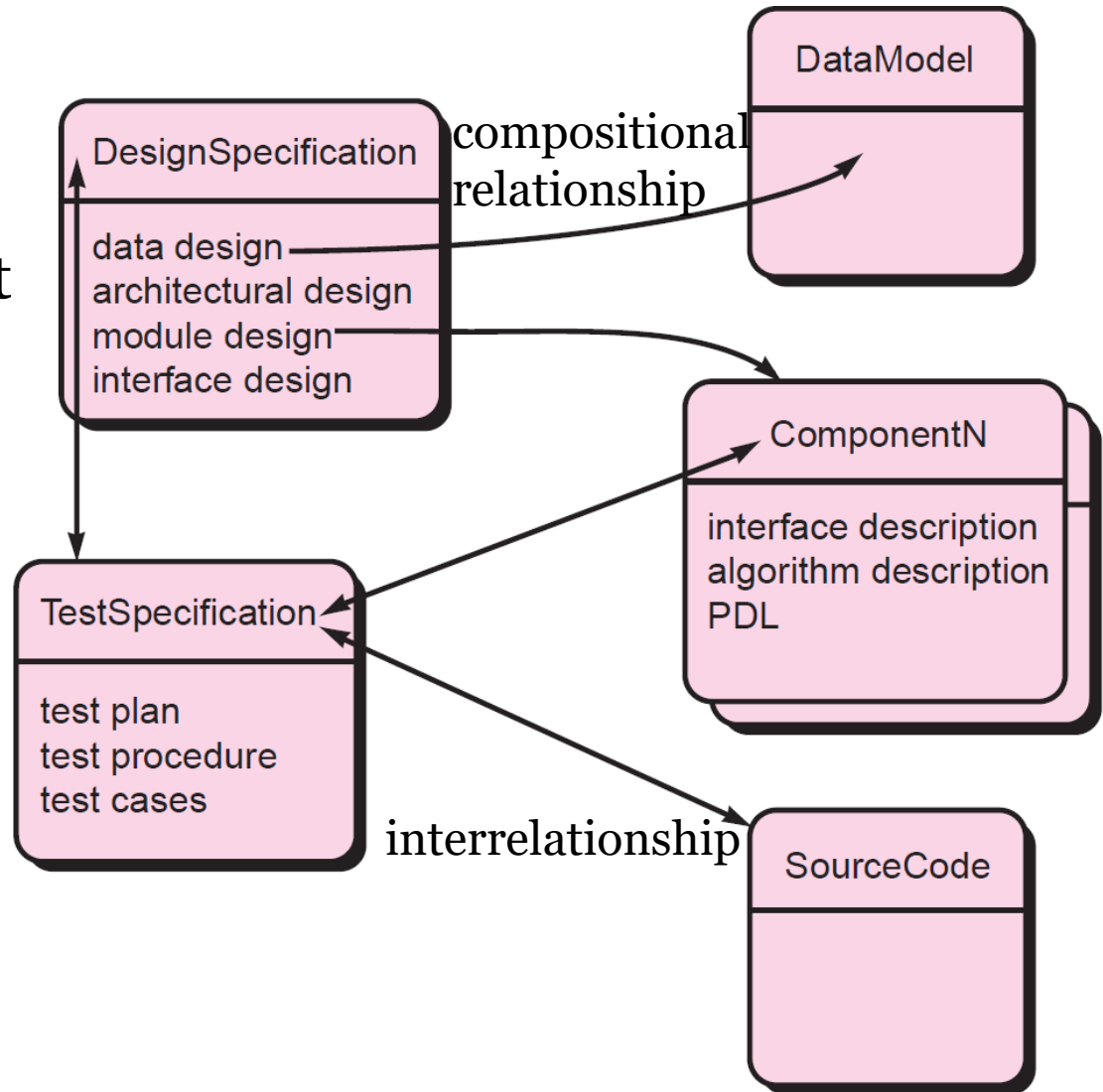
- The IEEE (IEEE Std. No. 610.12-1990) defines a baseline as:
 - A specification or product that has been formally reviewed and agreed upon, that thereafter serves as the basis for further development, and that can be changed only through formal change control procedures.
- A baseline is a milestone in the development of software that is marked by the delivery of one or more software configuration items (SCIs) and the approval of these SCIs that is obtained through a formal technical review

Baselines



Software Configuration Objects

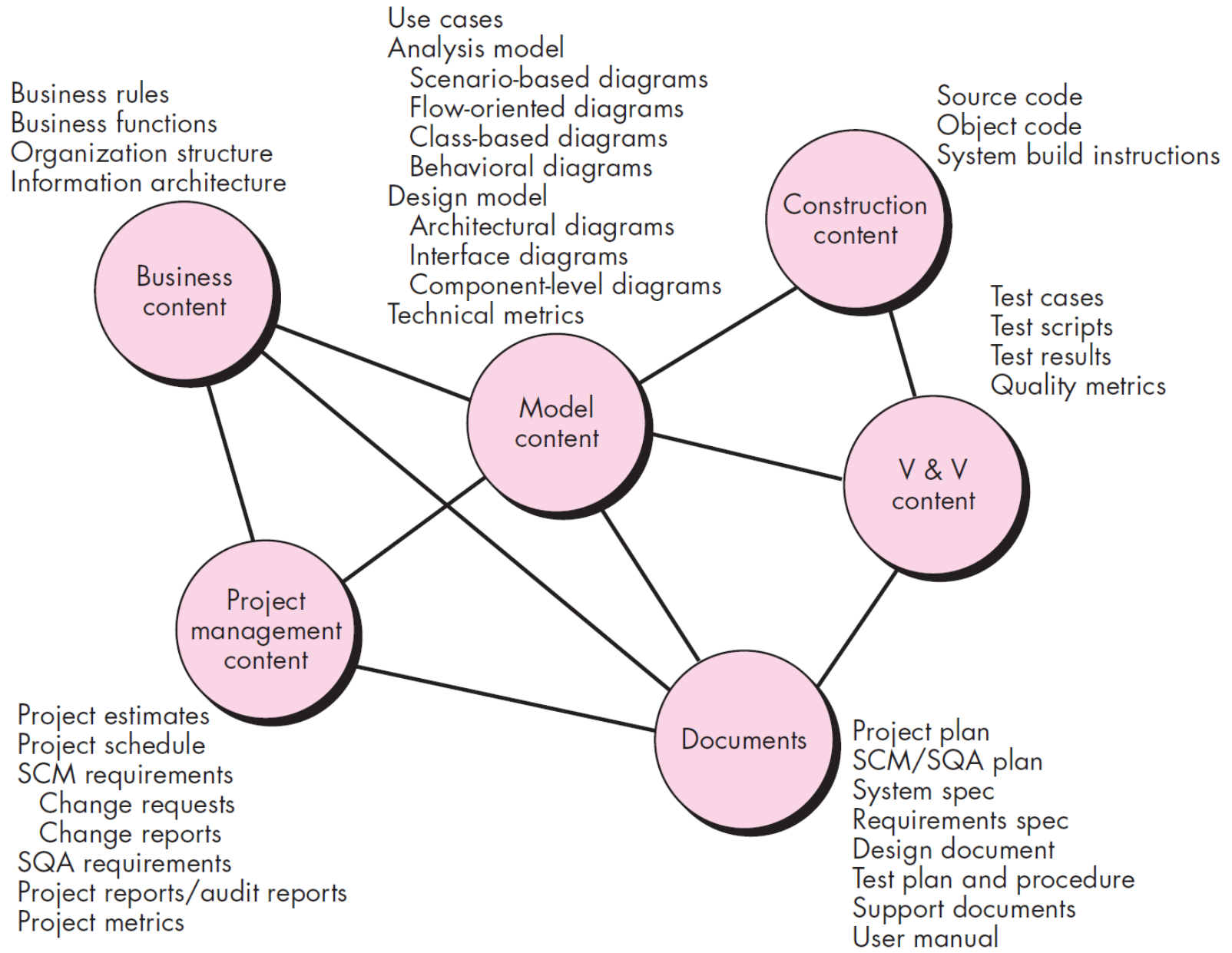
- SCIs are organized to form configuration objects that may be cataloged in the project database with a single name.
- A **configuration object** has a name, attributes, and is “connected” to other objects by relationships.



SCM Repository

- The SCM repository is the set of mechanisms and data structures that allow a software team to manage change in an effective manner (similar to a DBMS but also provides services specific to the software engineering environment)
- The repository performs or precipitates the following functions [For89]:
 - Data integrity
 - Information sharing
 - Tool integration
 - Data integration
 - Methodology enforcement
 - Document standardization

Repository Content



Repository Features

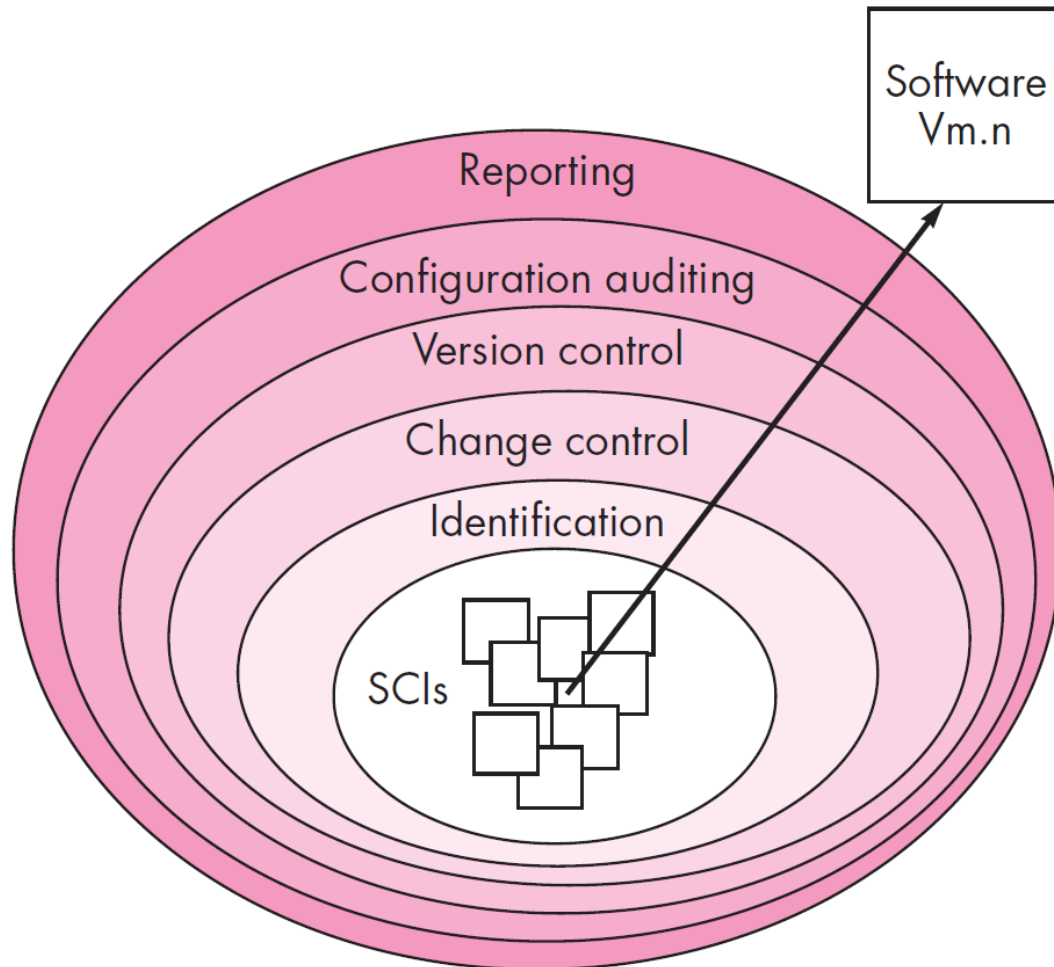
- **Versioning.**
 - Saves the versions to enable effective management of product releases and to permit developers to go back to previous versions
- **Dependency tracking and change management.**
 - The repository manages a wide variety of relationships among the data elements stored in it.
- **Requirements tracing.**
 - Provides the ability to track all the design and construction components and deliverables that result from a specific requirement specification
- **Configuration management.**
 - Keeps track of a series of configurations representing specific project milestones or production releases. Version management provides the needed versions, and link management keeps track of interdependencies.
- **Audit trails.**
 - Establishes additional information about when, why, and by whom changes are made.

The SCM Process

- Addresses the following questions ...
 - How does a software team identify the discrete elements of a software configuration?
 - How does an organization manage the many existing versions of a program (and its documentation) in a manner that will enable change to be accommodated efficiently?
 - How does an organization control changes before and after software is released to a customer?
 - Who has responsibility for approving and ranking changes?
 - How can we ensure that changes have been made properly?
 - What mechanism is used to inform others of changes that are made?

Layers of the SCM Process

- SCIs flow outward through the layers throughout their useful life, ultimately becoming part of the software configuration of one or more versions of an application or system.



Identification of objects

- To control and manage configuration items, each must be named and managed using an object-oriented approach
- Basic objects are created by software engineers during analysis, design, coding, or testing
- Aggregate objects are collections of basic objects and other aggregate objects
- Configuration object attributes: unique name, description, list of resources, and a realization (a pointer to a work product for a basic object or null for an aggregate object)

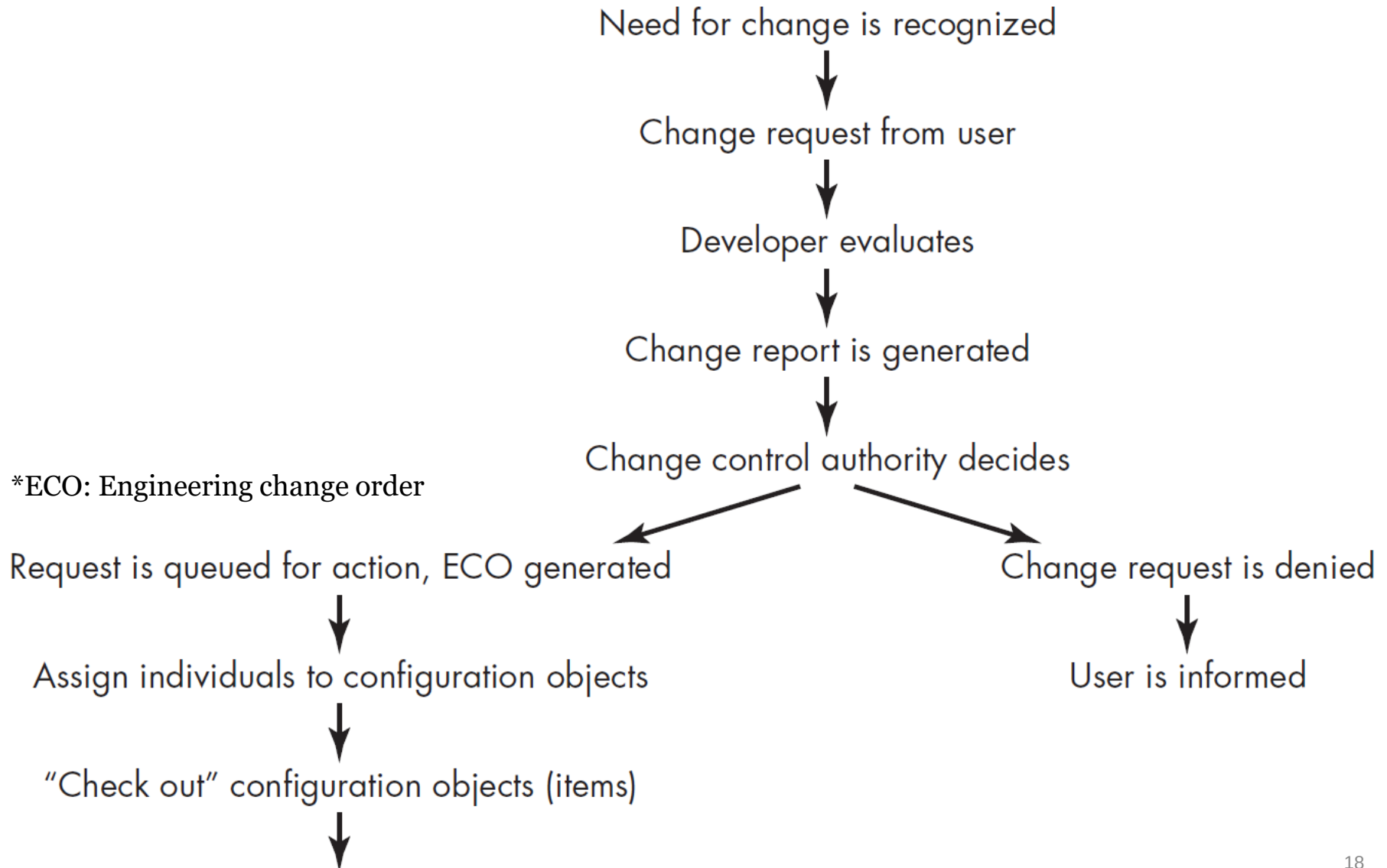
Version Control

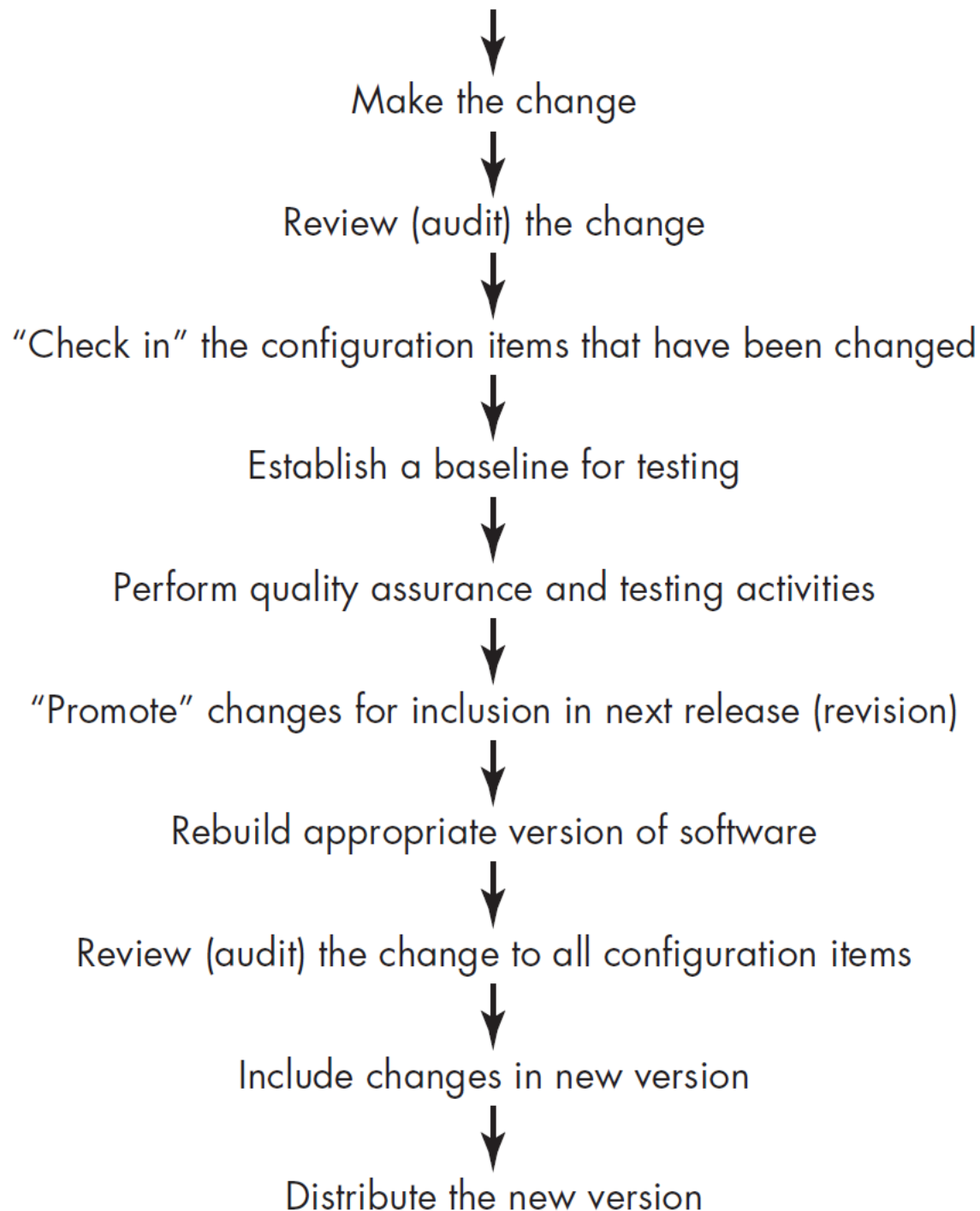
- Version control combines procedures and tools to manage different versions of configuration objects that are created during the software process
- A version control system implements or is directly integrated with four major capabilities:
 - a *project database (repository)* that stores all relevant configuration objects
 - a *version management* capability that stores all versions of a configuration object (or enables any version to be constructed using differences from past versions);
 - a *make facility* that enables the software engineer to collect all relevant configuration objects and construct a specific version of the software.
 - an *issues tracking* (also called bug tracking) capability that enables the team to record and track the status of all outstanding issues associated with each configuration object.

Change Control

- Change request is submitted and evaluated to assess technical merit and impact on the other configuration objects and budget
- Change report contains the results of the evaluation
- Change control authority (CCA) makes the final decision on the status and priority of the change based on the change report
- Engineering change order (ECO) is generated for each change approved (describes change, lists the constraints, and criteria for review and audit)
- Object to be changed is checked-out of the project database subject to access control parameters for the object
- Modified object is subjected to appropriate SQA and testing procedures
- Modified object is checked-in to the project database and version control mechanisms are used to create the next version of the software
- Synchronization control is used to ensure that parallel changes made by different people don't overwrite one another

Change Control Process



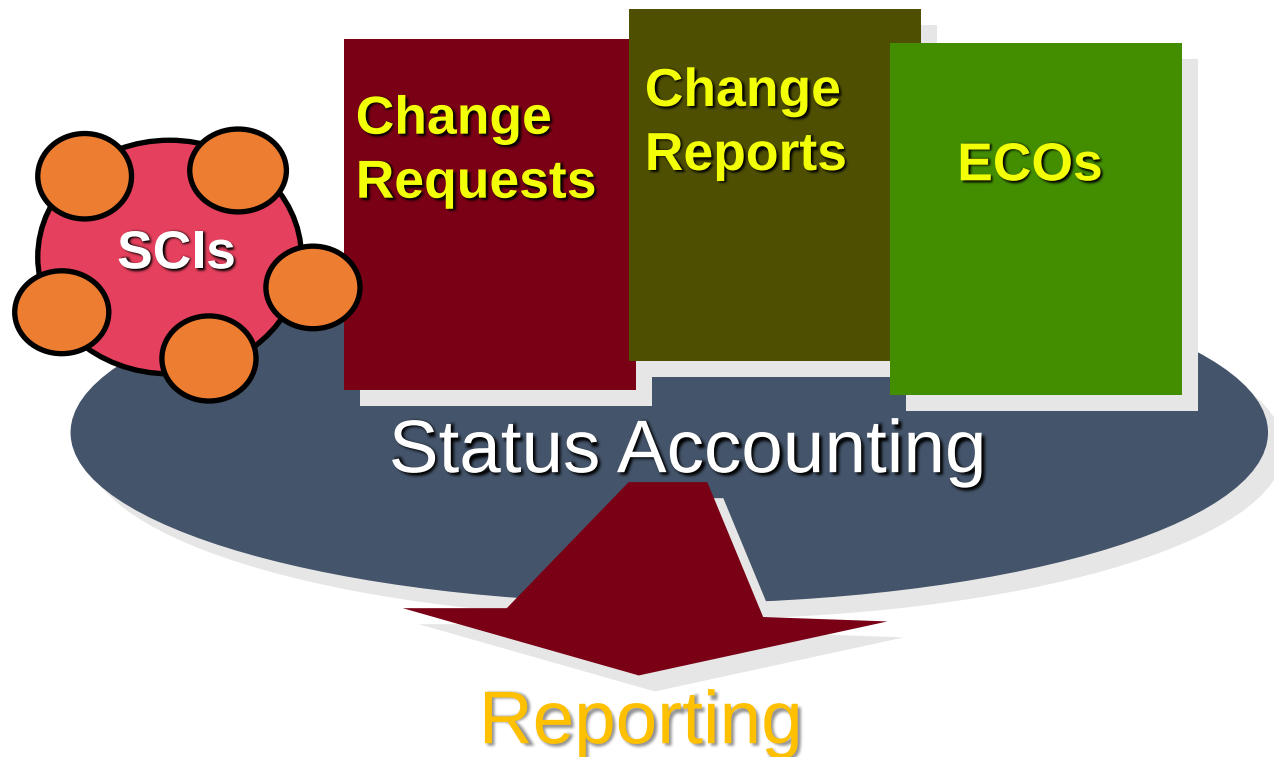


Configuration Audit

- How can a software team ensure that the change has been properly implemented?
 - A **technical review** focuses on the technical correctness of the configuration object that has been modified.
 - A **software configuration audit** complements the technical review by asking and answering the following questions:
 - Has the change specified in the ECO been made? Have any additional modifications been incorporated?
 - Has the software process been followed and have software engineering standards been properly applied?
 - Has the change been “highlighted” in the SCI? Have the change date and change author been specified?
 - Have SCM procedures for noting the change, recording it, and reporting it been followed?
 - Have all related SCIs been properly updated?

Configuration Status Reporting

- Sometimes called **status accounting**
- An SCM task that answers the following questions:
 - (1) What happened? (2) Who did it? (3) When did it happen? (4) What else will be affected?



Summary

- Software configuration management is an umbrella activity that is applied throughout the software process.
- SCM identifies, controls, audits, and reports modifications that invariably occur throughout the software life time.
- The software configuration is composed of a set of interrelated objects, i.e., software configuration items, that are produced as a result of some software engineering activity.
- Once a configuration object has been developed and reviewed, it becomes a baseline.
- Changes to a baselined object result in the creation of a new version of that object.
- Such changes are managed through the version control, change control, configuration audit, and status reporting layers of the SCM process.

CONFIGURATION MANAGEMENT TOOLS



Terraform

'SALTSTACK



puppet



git



CHEF™



ANSIBLE



docker